ALOK KUMAR SINGH

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CAREER OBJECTIVE:

High-impact Structural Simulation Engineer with 4+ years of experience delivering simulation-driven design solutions across defence, energy, and high-temperature mechanical systems. Expertise in advanced FEA, thermomechanical fatigue, nonlinear structural analysis, and Multiphysics coupling using ANSYS and Abaqus. Skilled in multi-body dynamics, fatigue, structural, modal and creep analysis for structural components under real-world and extreme load conditions. Seeking to apply simulation expertise and design leadership in a challenging R&D or structural simulation role focused on innovation and reliability.

CORE EXPERTISE

- **CAE & FEA Tools:** ANSYS (Workbench, Fluent, SpaceClaim, APDL Exposure), Abaqus, Basic Hypermesh
- **Analysis Types:** Static, Linear/Non-Linear, Fatigue (LCF/HCF), Modal, MBD, Random Vibration, Thermal-Structural, Steady-State Thermal, TMF, Creep and Basic NVH.
- **Automotive & Mechanical Systems:** Defence superstructures, Power plant headers, BFNV Wagons etc.
- **CAD Tools:** SolidWorks, AutoCAD
- **Automation and Scripting:** Python, MATLAB
- **Collaboration:** failure analysis, design reviews, technical documentation, stakeholder communication.

PROFESSIONAL EXPERIENCE

Sr. Project Engineer |IIT Bombay | Maharashtra | **May 2026 - Present**

- Create FEA model of gas tight premium thread connections.
- Performing nonlinear analysis in ANSYS to optimize the product design.
- Developing a comprehensive technical approach for CAL IV testing.

*(**Client:** ONGC, EOT & SDB group (Texas, USA))*

Co - Founder |Fortex Technology | Maharashtra | **Jul 2025 – May 2026**

- Co-founded an engineering simulation startup focused on FEA, CFD and Multiphysics consulting services.
- Developing simulation-driven engineering workflows for structural durability, thermal analysis and fluid systems.
- Building technical capabilities in nonlinear FEA, fatigue analysis and CFD-based design optimization.
- Engaging with industry partners and OEMs to establish simulation consulting collaborations.

(Startup Stage: Pre-revenue | Partnership development phase)

Design Engineer (R&D) |Technocraft Industries India Limited | Maharashtra | **June 2024 – May 2025**

- Created **detailed FE models** (**meshing, assembly, boundary conditions**) for defense structures using **ANSYS**.
- Performed **static, fatigue, and modal analyses** ensuring structural integrity under cyclic & dynamic loading.
- Conducted **CFD + structural coupling** for superstructure shutters; improved design safety margin by 12%.
- **Validated** CAE models against testing requirements and collaborated with cross-functional teams to propose countermeasures.
- Supported ₹100 Cr+ **defence tender projects** by delivering simulation-backed design insights that reduced approval cycles by 25%.

*(**Client:** DMDE, DRDO)*

Project Engineer | Indian institute of Technology| Bombay, Maharashtra | May 2023 – May 2024

- Performed **Thermomechanical Fatigue** (TMF) and **creep analysis** on thermal power plant components using **ABAQUS (Std. Code:** ASME BPVC VIII & ASME BPVC Section III - NH**)**
- Developed **3D FE models** of superheater headers; improved life prediction accuracy by 15% compared to legacy models.
- Delivered **fracture mechanics**-based **life predictions**, supporting Tata Power in extending service intervals. *(**Client:** Tata power company limited)*

Additional R&D Projects:

- Jay Chemicals Zero-Leak Dosing System: Contributed to **design optimization** and **prototype validation**, improving operational reliability (**Client:** Jay Chemicals)
- **Root cause Failure Analysis** of **BFNV Wagon (Client:** JSW**)**
  - Prepared a **detailed design proposal** for the **client** and conducted a comprehensive **literature review** to understand **design benchmarks** and **challenges**.
  - Performed preliminary **static structural simulations** to evaluate stress distribution and structural integrity.

CAE Intern| Kritsnam Technologies Pvt. Ltd.| Sept 2022 – April 2023

- Supported product R&D by performing **data analysis** and **prototype testing** for water flow monitoring systems.
- Assisted in evaluating **material and design durability** under operational loads, aligning with CAE-based performance assessments.
- Contributed to **sustainable design** improvements by combining experimental feedback with simulation-driven insights.

R&D Engineer | Terracent Materials| StartUp Nov 2020 – Feb 2022

- Conducted **literature reviews** on biodegradable plastics R&D and material applications.
- **Collaborated** with the startup **team** to identify sustainable material properties for development.

ACADEMIC QUALIFICATION

- M.Tech in Thermal Engineering | ZHCET, AMU | 2020 | CPI: 8.0 (First Division)
- B.E. Mechanical Engineering | ZHCET, AMU | 2017 | CPI: 8.81 (Honors, First Division)

PUBLICATIONS:

- ***Book Chapter:*** Experimental Study of Thermal Contact Conductance with Interstitial Materials, Springer (KIIT Thermo 2020).
- ***Conference:*** Enhancement of Thermal Contact Conductance, ISHMT-ASTFE Heat & Mass Transfer Conference.
- ***Journal:*** Experimental Study of Thermal Contacts for Interfacial Heat Transfer Enhancement, *Engineering Research Express*.

CERTIFICATES:

- Istec vibration Training Category 1+, 2025
- ANSYS Learning Track: Foundation of Heat transfer in structures
- ANSYS Course Completion: Multi-Die Semiconductor Integrity
- MATLAB/SIMULINK Bible| go From Zero to Hero **Link**
- Responsive Web Design, freeCodeCamp, 2024 **Link**